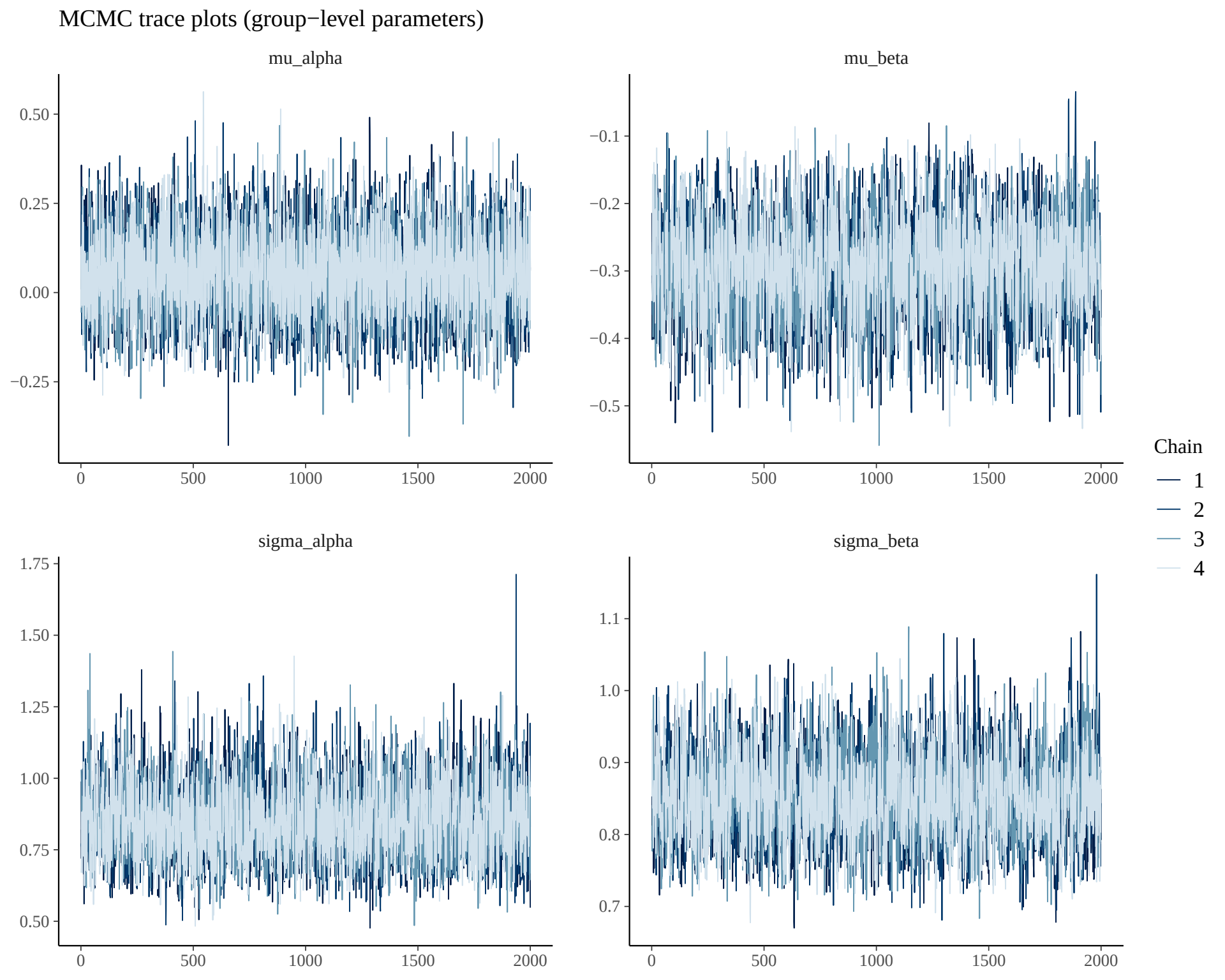
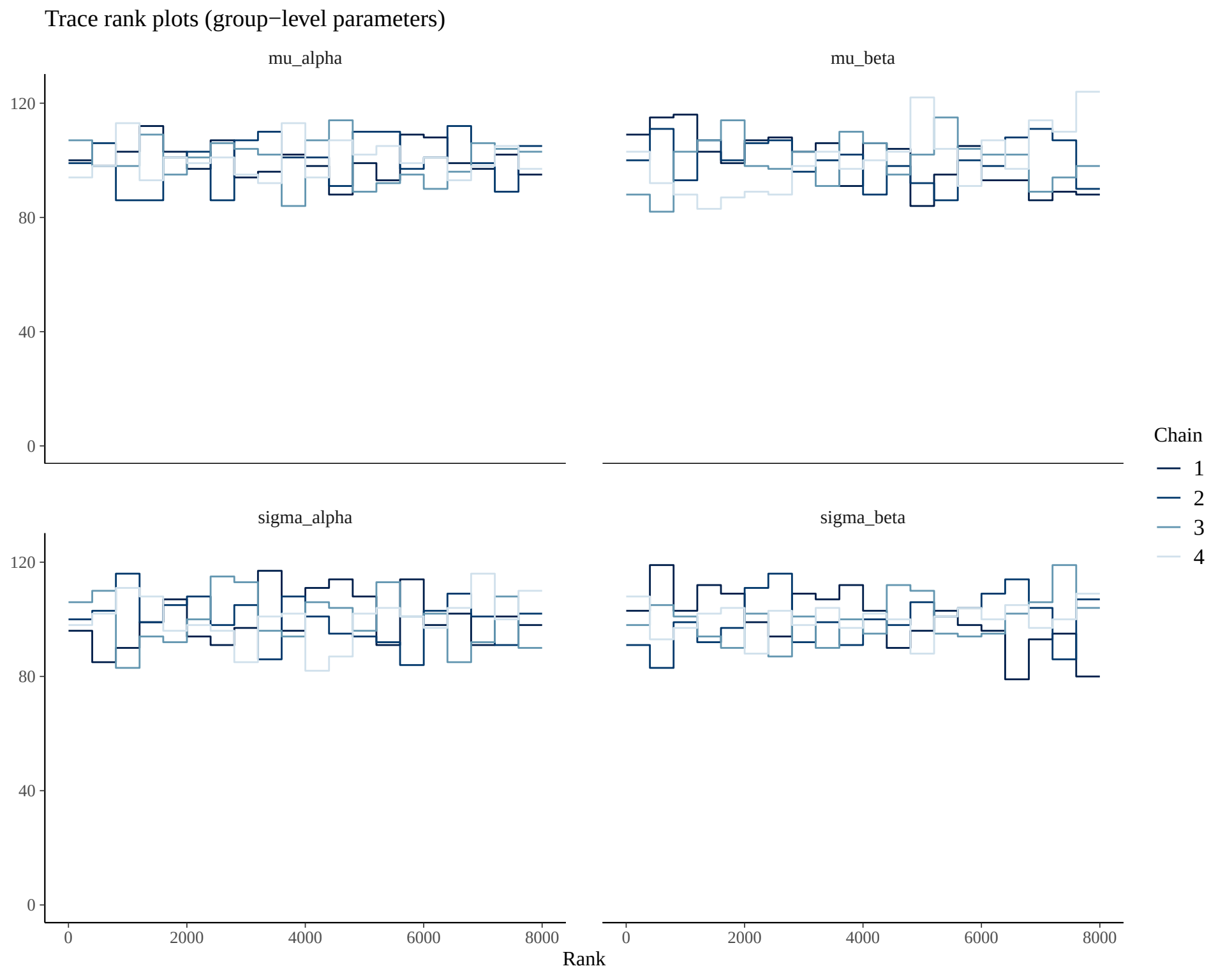
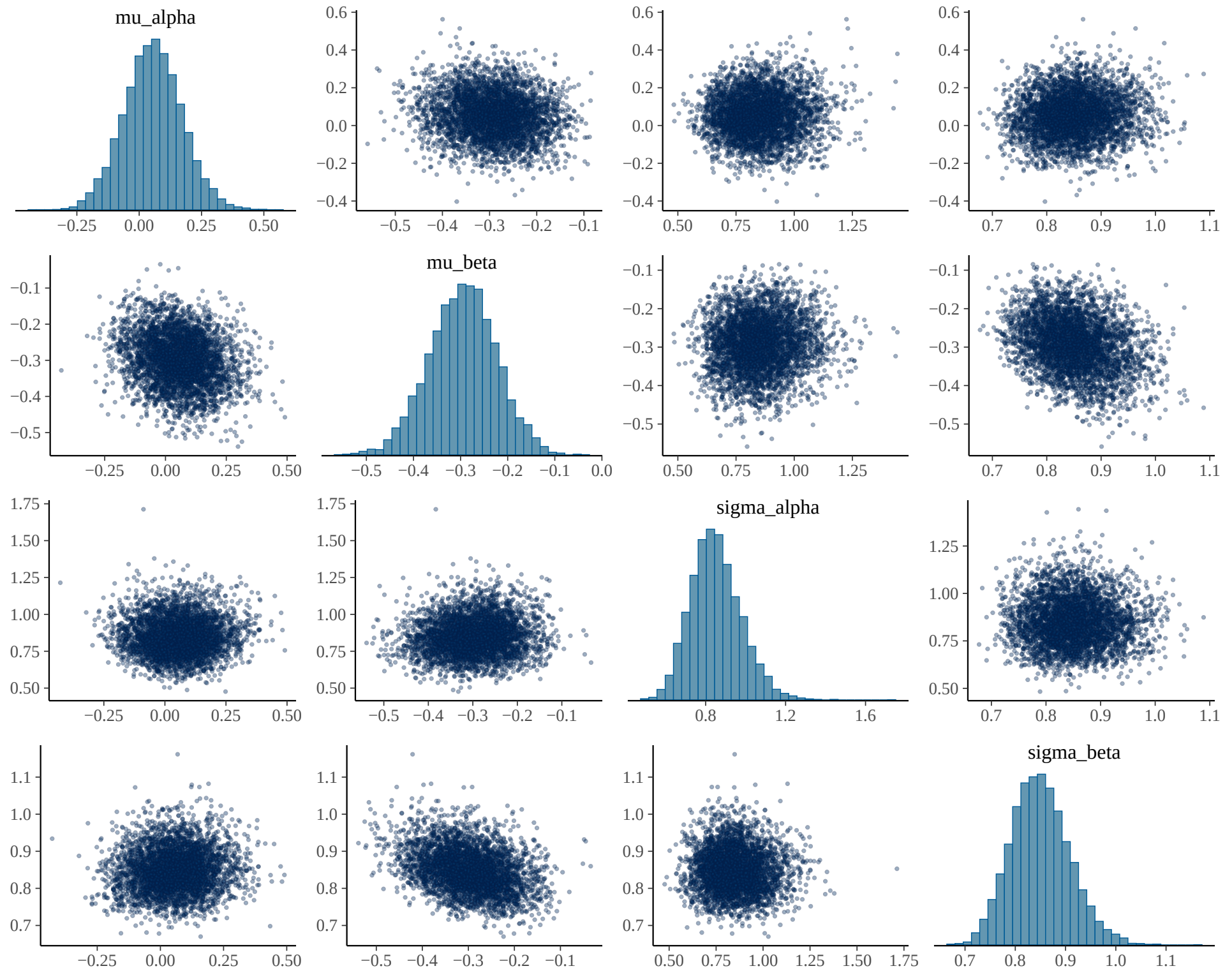
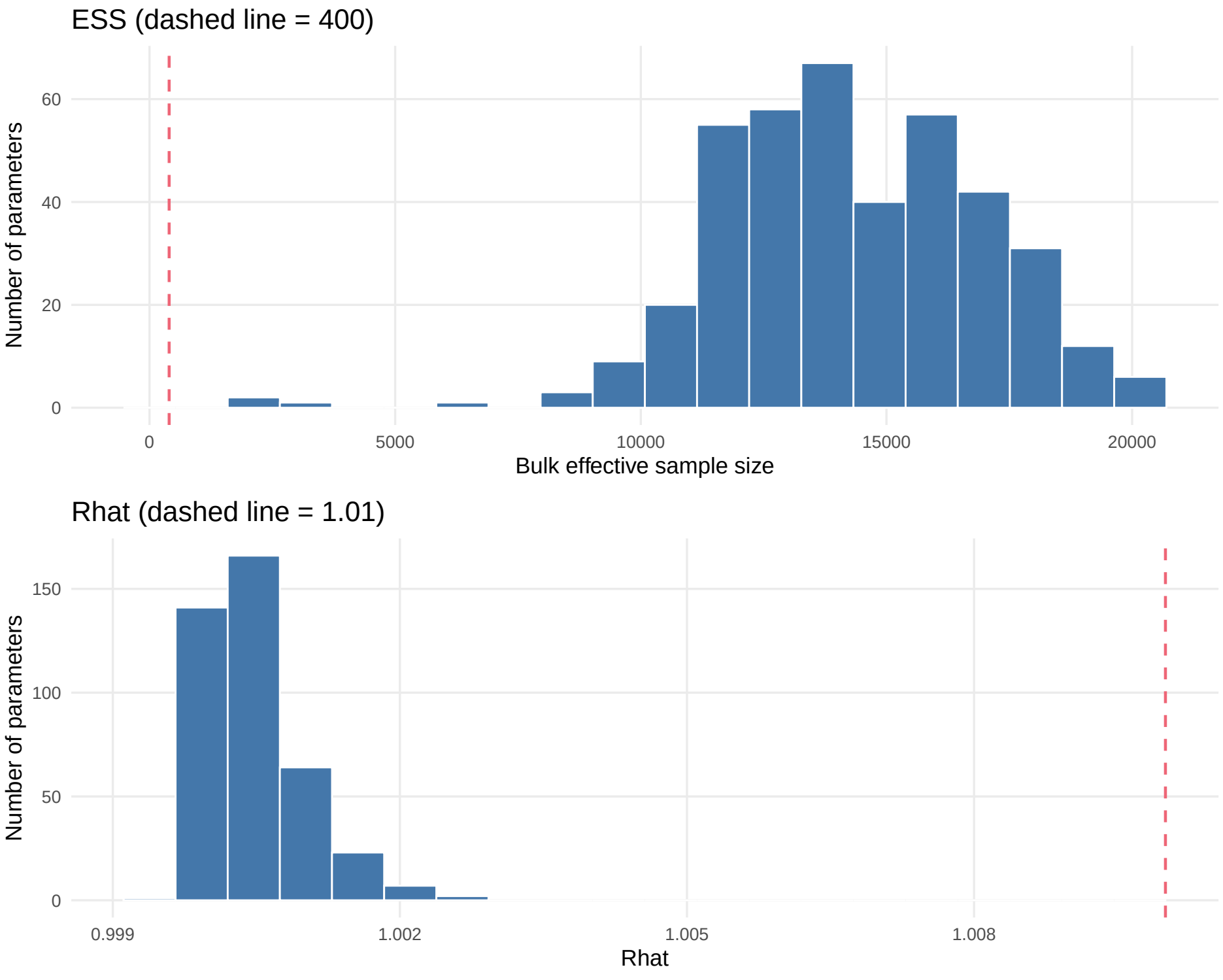

MCMC trace plots (group−level parameters)
mu_alpha
mu_beta
sigma_alpha
sigma_beta
0.50
0.25
0.00
−0.25
−0.1
−0.2
−0.3
−0.4
−0.5
1.75
1.50
1.25
1.00
0.75
0.50
1.1
1.0
0.9
0.8
0.7
0
500
1000
1500
2000
Chain
1
2
3
4


Trace rank plots (group−level parameters)
mu_alpha
mu_beta
sigma_alpha
sigma_beta
120
80
40
0
0
2000
4000
6000
8000
Rank
Chain
1
2
3
4


mu_alpha
mu_beta
sigma_alpha
sigma_beta


ESS (dashed line = 400)
Number of parameters
60
40
20
0
0
5000
10000
15000
20000
Bulk effective sample size
Rhat (dashed line = 1.01)
Number of parameters
150
100
50
0
0.999
1.002
1.005
1.008
Rhat